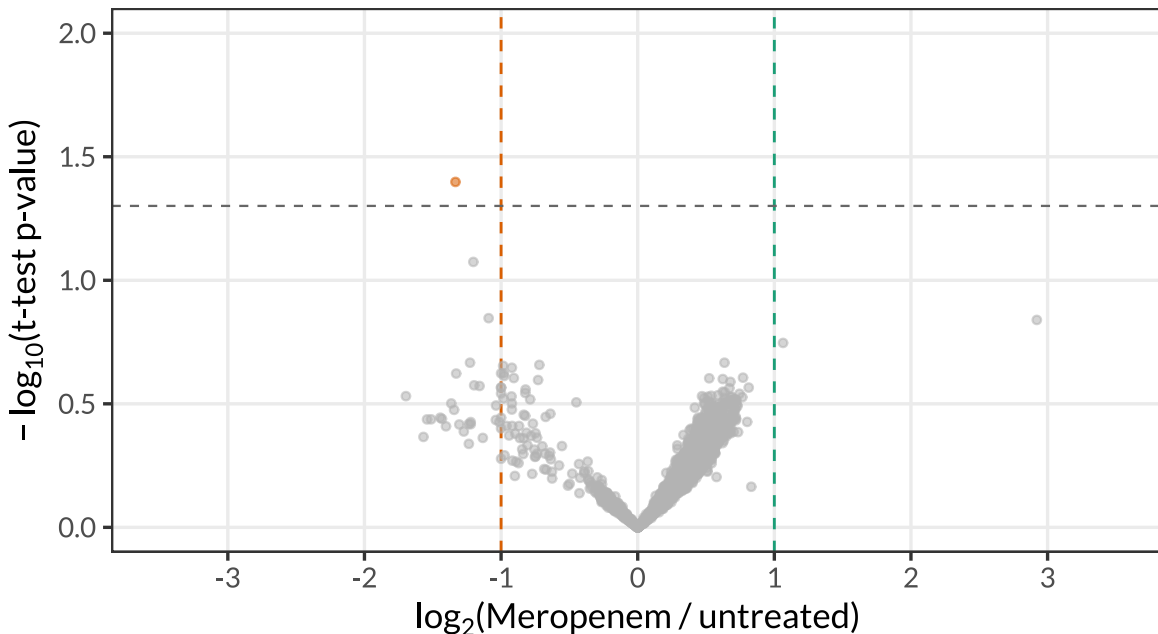


Meropenem vs untreated: volcano plot



Significance (t-test) ● Down (Meropenem-depleted) ● Not significant

Strain ATCC 17978-VU, Stationary phase. Panel C: volcano plot, $\log_2(\text{Meropenem}/\text{untreated})$ per 1kb window (x) vs $-\log_{10}(\text{t-test p-value})$ (y); $n = 4004$ windows (0 excluded for zero coverage in at least one condition); 0 significantly up, 1 significantly down ($|\log_2\text{FC}| > 1$ and t-test $p < 0.05$). Vertical lines mark $\log_2\text{FC} = -1/+1$; horizontal line marks $p = 0.05$. Caution: p-values are per-window from only 2 replicates per condition (t-test $\text{df} = 2$); BH-adjusted p-values are in the results CSV. Mann-Whitney U p-values (also in the CSV) are of limited value at this replicate count: the exact test's minimum attainable two-sided p-value is $1/3$ with no ties among the four values, but read counts are discrete and frequently tied, which makes R fall back to a normal approximation that can occasionally read below that nominal floor. Either way it is under-powered here and is not used to drive the y-axis (0 of 4004 windows show raw Mann-Whitney U $p < 0.05$, for reference).